

Data Analytics -- Guided Notes ANSWER KEY

Unit tp-5.3 | CompTIA Network+ N10-009 Obj. FC0-U71 5.3 | N10-008 FC0-U71 5.3

For instructor use / distribute after students complete the notes.

Question 1

Data analytics is the process of examining _____ to discover useful information and support decision-making. The four types of analytics in order of complexity are: _____ (what happened), _____ (why it happened), _____ (what will happen), and _____ (what should we do about it).

Answer: Data (datasets); descriptive; diagnostic; predictive; prescriptive.

Question 2

The _____ is the central repository that aggregates data from multiple sources for reporting and analysis. Unlike an operational database (optimized for _____ transactions), a data warehouse is optimized for _____ analytical queries across large historical datasets.

Answer: Data warehouse; fast read/write; complex (read-heavy).

Question 3

ETL stands for _____, _____, and _____. It is the process of moving data from source systems into a data warehouse. _____ pulls data from source systems. _____ cleans and reformats it into a consistent structure. _____ loads it into the destination.

Answer: Extract, Transform, Load; Extract; Transform; Load.

Question 4

Data _____ is the practice of examining large datasets to discover patterns, correlations, and anomalies. It uses techniques from statistics and machine learning. Business applications include _____ (flagging unusual transactions), _____ (anticipating what customers will buy), and _____ (finding patterns that predict equipment failure).

Answer: Mining; fraud detection; recommendation engines; predictive maintenance.

Question 5

_____ is data that is too large, too fast, or too varied for traditional databases to handle. The three Vs of big data are: _____ (massive amounts), _____ (high speed of generation), and _____ (multiple formats and sources). Technologies like Hadoop and Apache Spark process big data across _____ of servers in parallel.

Answer: Big data; volume; velocity; variety; clusters.

Question 6

A _____ is a visual representation of data designed to communicate patterns and insights clearly. Common chart types: _____ chart for comparing categories, _____ chart for parts of a whole, _____ chart for trends over time, and _____ plot for showing correlation between two variables.

Answer: Data visualization; bar; pie; line; scatter.

Question 7

_____ are commonly used measures in data analysis. The _____ is the sum divided by the count. The _____ is the middle value when sorted. The _____ is the most frequently occurring value. The _____ is the difference between the maximum and minimum values.

Answer: Descriptive statistics; mean (average); median; mode; range.

Question 8 -- Real World Application

REAL WORLD: A school cafeteria wants to reduce food waste. The food services director collects 6 months of daily meal sales data. Describe which type of analytics they should use for each of the following goals: (a) understand which meals are most popular, (b) understand why Monday waste is 40% higher than other days, (c) predict how many portions of each meal to prepare next week.

Answer: (a) Descriptive analytics -- analyze historical sales counts per meal item to see what was most purchased. Tools: aggregate counts, bar charts, ranking tables. This answers 'what happened.' (b) *Diagnostic analytics -- drill into Monday data to find the why. Compare Monday menus vs. other days, check if Monday attendance is lower, examine if certain unpopular meals are scheduled Mondays. This answers 'why did it happen.'* (c) *Predictive analytics -- use historical demand patterns, day-of-week trends, and seasonal patterns (e.g., spike after school breaks) to build a forecast model that predicts portions needed per meal next week. This answers 'what will happen.'* Tools might include a simple regression model in Excel or a machine learning model in Python.